

Lotrène® FE8000

LOW DENSITY POLYETHYLENE
(LDPE)

DESCRIPTION

Lotrène® FE8000 is an additive free grade, produced by a high pressure tubular process, mainly recommended for light and medium duty film applications with good optical properties and medium rigidity.

PROPERTIES

Films produced with Lotrène® FE8000 are sufficiently rigid for automatic packaging machines, suitable for shrink wrapping and have good optical properties.

PROCESSING

Lotrène® FE8000 can be easily processed on all types of extruders designed for polyethylene and over a wide range of extrusion conditions.

The melt temperature is suggested to be in the range of 160-180 °C.

The best properties of the blown film are achieved at blow up ratios between 2:1 and 3:1.

The recommended thickness range is from 30 µm to 200 µm.

POLYMER PROPERTIES	VALUE	UNIT	TEST METHOD
Melt flow index	0.80	g/10 min.	ASTM D-1238
Density @ 23 °C	0.923	g/cm³	ASTM D-792
Crystalline melting point	111	°C	ASTM E-794
Vicat softening point	97	°C	ASTM D-1525

FILM PROPERTIES	VALUE	UNIT	TEST METHOD
Tensile strength @ yield MD/ TD	12/12	MPa	ASTM D-882
Tensile strength @ break MD/ TD	34/27	MPa	ASTM D-882
Elongation @ break MD/ TD	410/615	%	ASTM D-882
Impact strength, F ₅₀	115	g	ASTM D-1709
Tear resistance MD/TD	70/40	N/mm	ASTM D- 1922
Puncture force	51	N	Internal Method
Coefficient of friction	0.9		ASTM D-1894
Haze	7	%	ASTM D-1003
Gloss @ 45°	61		ASTM D-2457

Film properties stated above have been obtained using 40 µm blown films laboratory test specimens produced under following conditions: 45 mm screw with L/D = 30, die diameter 120 mm, die gap 1.56 mm, BUR 2:5:1

APPLICATIONS

- Shrink wrap film for light and medium duties
- Shopping bags
- Lamination film

HANDLING & STORAGE

Polyethylene products should be stored in their original packaging or in clean appropriate silos. The products should be stored in a dry and well-ventilated area and should not be exposed to direct sunlight and/or heat in any form since this may adversely affect their properties.

As a general rule, our products should not be stored for more than three months from receipt date.

SAFETY

Under normal conditions Lotrène® products do not present a toxic hazard through skin contact or inhalation. For detailed information please refer to the Safety Data Sheet.

FOOD CONTACT & REACH

Lotrène® polyethylene products manufactured by Qatar Petrochemical Company (QAPCO) Q.P.J.S.C. comply with US, EU and other food contact legislations. Limitations may apply. Please contact your QatarEnergy marketing representative for detailed compliance certificates.

All QAPCO Lotrène® products are complying with REACH Regulation 1907/2006/EC. The aims of this regulation are to improve the protection of human health and the environment through better and earlier identification of the intrinsic properties of chemical substances.

NOT SUITABLE FOR PHARMACEUTICAL OR MEDICAL APPLICATIONS

Lotrène® products are not suitable for pharmaceutical or medical applications.

TECHNICAL DISCLAIMER

Before using this product, the user is advised and cautioned to make its own determination and assessment of the safety and suitability of the product for the specific use in question and is further advised against relying on the information contained herein as it may relate to any specific use or application. It is the ultimate responsibility of the user to ensure that the product is suited for and the information is applicable to the user's specific application. Any data included herein is based upon analysis of representative samples in a laboratory environment and not the actual product shipped. The values indicated in this datasheet should not be used for specification purposes.

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